

Solving Systems of Linear Equations

Elimination · Matrix row reduction · Row operations that preserve singularity · Rank of a matrix ·
Row echelon form · Reduced row echelon form · Gaussian elimination

Roadmap for this week

1 Solving systems of equations

Non-singular, singular, and systems with more variables.

2 Matrix row reduction

Turning a system into a matrix and reducing its rows.

3 Row operations that preserve singularity

Swap, scale, add, and what happens to the determinant.

4 Rank of a matrix

How much information a system really carries.

5 Row echelon form and reduced row echelon form

Pivots, the staircase, and counting the rank.

6 Gaussian elimination

The complete algorithm on the augmented matrix.

We start with the machine-learning motivation. Three short in-class quizzes follow: after non-singular systems, after singular systems, and after rank.

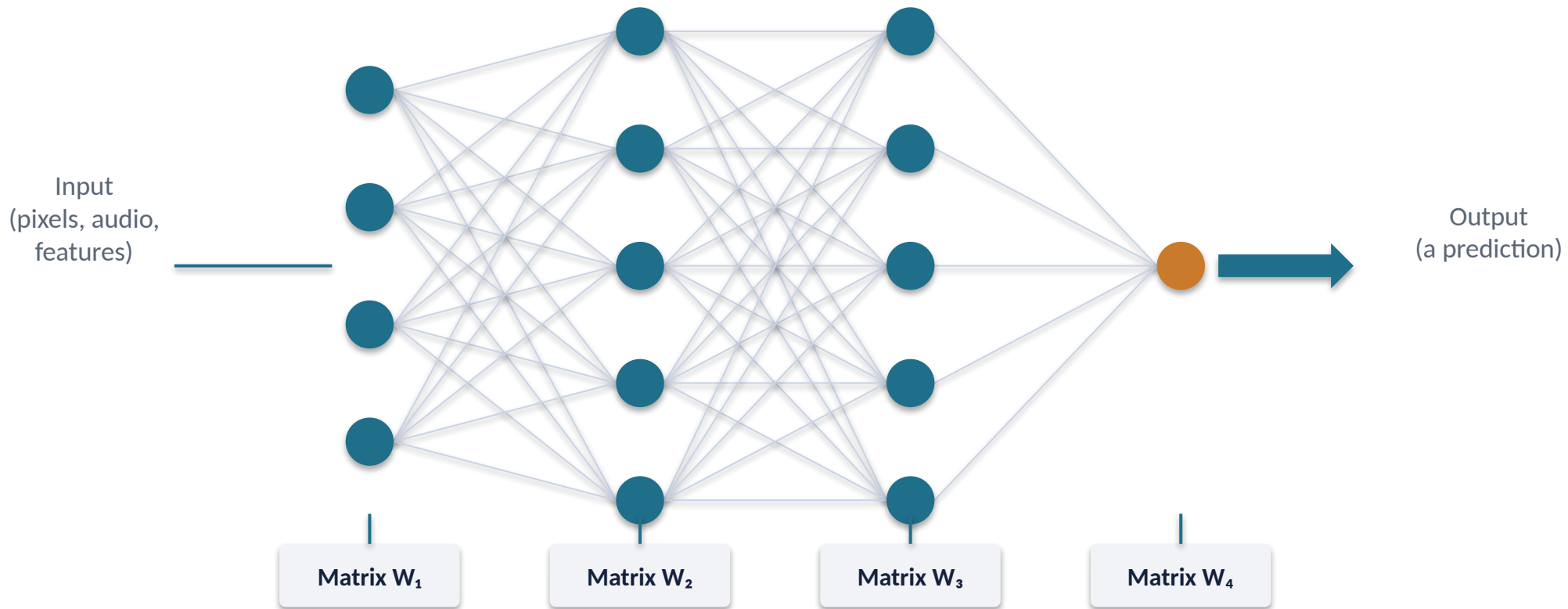
MOTIVATION

Why systems of linear equations matter in machine learning

Behind every neural network, regression model and recommender system sits a matrix. This week is about what those matrices tell us and how to solve for them.

Neural networks are matrix operations

A neural network is a chain of layers. Each layer multiplies its input by a matrix of weights and adds a bias. Running the network is matrix multiplication; training it is solving for the entries of those matrices.



Everything this week (solving systems, row reduction, rank) is the toolkit for reasoning about these matrices.

Where this shows up in data science

ACOUSTIC MONITORING

Sound recognition

Underwater microphones record the sounds of fish and marine mammals; a neural network classifies the species. Tracking species through sound helps preserve bio-habitats. The audio becomes a matrix of features, and the network is a stack of matrix operations.

GENERATIVE MODELS

AI-generated music

Automatic music generation compresses music into discrete codes, then trains a model on a specific genre to produce new pieces. Learning the compression and the generator means fitting very large matrices.

THE SIMPLEST MODEL

Linear regression

Fitting a line (or a hyperplane) to data means solving the normal equations $X^T X \beta = X^T y$: a system of linear equations in the coefficients β . Whether it has one solution, many, or none is exactly today's question.

Same mathematics in each case: data in a matrix, unknowns in a vector, equations linking them.

PART 1

Solving systems of linear equations

From two equations in two unknowns to three equations in three unknowns, using only three legal moves.

Linear systems

DEFINITION

A **linear equation** in the variables $a, b, c, \dots$ is an equation of the form

$$\alpha_1 a + \alpha_2 b + \alpha_3 c + \dots = \beta,$$

where the coefficients $\alpha_1, \alpha_2, \dots$ and the constant β are given numbers. No products or powers of the variables appear.

A **system of linear equations** is a finite collection of linear equations in the same variables, considered together.

A **solution** of the system is an assignment of numbers to the variables that satisfies every equation at the same time.

EXAMPLES

$$a + b = 10$$

$$a + 2b = 12$$

two equations, two unknowns; the solution is $a = 8, b = 2$

$$a + b + 2c = 12$$

$$3a - 3b - c = 3$$

$$2a - b + 6c = 24$$

three equations, three unknowns; we solve it later today

DATA SCIENCE CONNECTION

Fitting a linear regression means solving the normal equations $X^T X \beta = X^T y$, a system of exactly this kind.

A system and its solution

An apple and a banana cost \$10. An apple and two bananas cost \$12. What does each cost?

SYSTEM

$$a + b = 10$$

$$a + 2b = 12$$

Some process:
manipulating the equations



swapping equations
adding equations
multiplying an equation by a constant

SOLVED SYSTEM

$$a = 8$$

$$b = 2$$

Reading it off: the second purchase differs from the first by exactly one banana, so a banana costs $12 - 10 = 2$, and the apple costs $10 - 2 = 8$. Elimination turns this intuition into a procedure.

Equivalent systems and the three legal operations

DEFINITION

Two systems are **equivalent** if they have exactly the same solutions. Each of the following operations replaces a system by an equivalent one:

(1) swap two equations; **(2)** multiply an equation by a non-zero constant; **(3)** add a multiple of one equation to another.

Solving a system means applying these operations until each equation involves a single variable.

MULTIPLYING BY A CONSTANT

$$a + b = 10$$

$$\times 7$$

$$\mathbf{7a + 7b = 70}$$

ADDING TWO EQUATIONS

$$a + b = 10$$

$$+ (2a + 3b = 22)$$

$$\mathbf{3a + 4b = 32}$$

Worked example: two equations, two unknowns

1

System

$$5a + b = 17$$

$$4a - 3b = 6$$

2

Divide by the coefficient
of a

$$a + 0.2b = 3.4$$

$$a - 0.75b = 1.5$$

Both equations now start with a.

3

Subtract equation 1 from
equation 2

$$-0.95b = -1.9$$

$$\mathbf{b = 2}$$

a is eliminated from equation 2.

4

Back-substitute $b = 2$

$$a + 0.2(2) = 3.4$$

$$\mathbf{a = 3}$$

Solution: $a = 3, b = 2$.

The pattern: make the leading coefficients equal, subtract to eliminate a variable, solve the smaller system, then substitute back. Every step is one of the three legal operations, so the final system is equivalent to the original.

What if a coefficient of a is already zero?

SYSTEM

$$5a + b = 17$$

$$3b = 6$$



ISOLATE b

$$3b = 6$$

$$\mathbf{b = 2}$$



SOLVED

$$a = 3$$

$$b = 2$$

There is nothing to eliminate: a already has coefficient 0 in the second equation, so that equation gives b directly.

Then $a + 0.2(2) = 3.4$ gives $a + 0.4 = 3.4$, so $a = 3$.

In matrix language this zero is a head start: the row is already in the shape elimination is trying to produce.

Solve the following system of equations

$$2a + 5b = 46$$

$$8a + b = 32$$

Use the three legal operations: scale an equation, subtract to eliminate one variable, then back-substitute.

Write your answer as a pair (a, b).

Quiz 1 · Solution

SYSTEM

$$2a + 5b = 46$$

$$8a + b = 32$$

SOLUTION

$$\mathbf{a = 3, b = 8}$$

ELIMINATE b

Multiply equation 2 by 5: $40a + 5b = 160$

Subtract equation 1: $38a = 114$

$$\mathbf{a = 3}$$

BACK-SUBSTITUTE

$$8(3) + b = 32$$

$$\mathbf{b = 8}$$

CHECK

$$2(3) + 5(8) = 6 + 40 = 46 \quad \checkmark$$

$$8(3) + 8 = 24 + 8 = 32 \quad \checkmark$$

What if the system is singular? (redundant)

SYSTEM

$$a + b = 10$$

$$2a + 2b = 20$$



divide eq. 2 by 2

$$a + b = 10$$

$$a + b = 10$$



subtract eq. 1

$$a + b = 10$$

$$0 = 0$$

The second equation carried no new information.

After elimination only one equation survives, and one equation cannot pin down two unknowns.

Pick any value $a = x$. Then $b = 10 - x$ also works: the system has infinitely many solutions, with one degree of freedom.

SOLUTION SET

$$a = x$$

$$b = 10 - x$$

for every real number x

What if the system is singular? (contradictory)

SYSTEM

$$a + b = 10$$

$$2a + 2b = 24$$



divide eq. 2 by 2

$$a + b = 10$$

$$a + b = 12$$



subtract eq. 1

$$a + b = 10$$

$$0 = 2$$

A contradiction.

No choice of a and b can make 0 equal to 2, so no pair satisfies both equations.

The left-hand sides are proportional ($2a + 2b$ is twice $a + b$) but the right-hand sides are not (24 is not twice 10).

SOLUTION SET

empty

the system has no solutions

Classifying systems

DEFINITION

A system is **consistent** if it has at least one solution and **inconsistent** if it has none.

A square system (as many equations as unknowns) is **non-singular** if it has exactly one solution, and **singular** otherwise, that is, when it has infinitely many solutions or none.

Unique solution

NON-SINGULAR

$$a + b = 10$$

$$a + 2b = 12$$

Geometrically: two lines that cross in one point

Infinitely many solutions

SINGULAR · REDUNDANT

$$a + b = 10$$

$$2a + 2b = 20$$

Geometrically: two equations for the same line

No solution

SINGULAR · CONTRADICTORY

$$a + b = 10$$

$$2a + 2b = 24$$

Geometrically: two parallel lines

Solve the following system of equations

$$5a + b = 11$$

$$10a + 2b = 22$$

Look closely at the two equations before you start computing.

Is the system non-singular, redundant, or contradictory? Describe every solution.

Quiz 2 · Solution

SYSTEM

$$5a + b = 11$$

$$10a + 2b = 22$$

SOLUTION SET

$$\mathbf{b = 11 - 5a}$$

for any real number a

ELIMINATE

Divide equation 2 by 2: $5a + b = 11$

This is equation 1 again. Subtracting gives

$$\mathbf{0 = 0}$$

The system is singular and redundant: equation 2 is just twice equation 1, so it adds no information.

One equation in two unknowns leaves one degree of freedom, so there are infinitely many solutions.

CHECK

$(a, b) = (0, 11), (1, 6), (2, 1), (-1, 16), \dots$ all satisfy both equations.

Elimination with three variables (1 of 3)

SYSTEM

$$a + b + 2c = 12$$

$$3a - 3b - c = 3$$

$$2a - b + 6c = 24$$

Three equations, three unknowns.

DIVIDE EACH ROW BY ITS COEFFICIENT OF a

$$a + b + 2c = 12$$

$$a - b - \frac{1}{3}c = 1$$

$$a - \frac{1}{2}b + 3c = 12$$

Now every equation starts with a.

SUBTRACT EQUATION 1 FROM THE OTHERS

$$a + b + 2c = 12$$

$$-2b - \frac{7}{3}c = -11$$

$$-\frac{3}{2}b + c = 0$$

a is isolated in equation 1; equations 2 and 3 form a 2×2 system in b and c.

Same idea as before: use the first equation to remove a from every other equation, then solve what is left.

Elimination with three variables (2 of 3)

THE SYSTEM SO FAR

$$a + b + 2c = 12$$

$$-2b - 7/3 c = -11$$

$$-3/2 b + c = 0$$

Work on the last two equations only.

DIVIDE BY THE COEFFICIENT OF b

$$a + b + 2c = 12$$

$$b + 7/6 c = 11/2$$

$$b - 2/3 c = 0$$

Both now start with b.

SUBTRACT EQUATION 2 FROM EQUATION 3

$$a + b + 2c = 12$$

$$b + 7/6 c = 11/2$$

$$-11/6 c = -11/2$$

b is isolated in equation 2, and the last equation gives $c = 3$.

The system is now triangular: the first equation has a, b, c; the second has b, c; the third has c alone.

Three variables (3 of 3): back substitution

TRIANGULAR SYSTEM

$$a + b + 2c = 12$$

$$b + 7/6 c = 11/2$$

$$c = 3$$

Back substitution: solve the last equation first and feed each answer into the equation above it.

SUBSTITUTE UPWARDS

$c = 3$ into equation 2: $b + 7/6 \cdot 3 = 11/2$, so $b + 7/2 = 11/2$, so $b = 2$

$b = 2, c = 3$ into equation 1: $a + 2 + 6 = 12$, so $a = 4$

Solution: $a = 4, b = 2, c = 3$

CHECK

$$4 + 2 + 2(3) = 12 \quad \checkmark$$

$$3(4) - 3(2) - 3 = 3 \quad \checkmark$$

$$2(4) - 2 + 6(3) = 24 \quad \checkmark$$

PART 2

Matrix row reduction

The same elimination steps, written on the coefficients only. Rows replace equations.

From systems to matrices

DEFINITION

The **coefficient matrix** of a system lists the coefficients of the variables: one row per equation, one column per variable. **Row reduction** applies the three legal operations to the rows of the matrix instead of to the equations.

ORIGINAL SYSTEM

$$5a + b = 17$$

$$4a - 3b = 6$$

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$$

Original matrix



INTERMEDIATE SYSTEM

$$a + 0.2b = 3.4$$

$$b = 2$$

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Upper triangular: row echelon form



SOLVED SYSTEM

$$1a + 0b = 3$$

$$0a + 1b = 2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Diagonal: reduced row echelon form

Singular systems as matrices

ORIGINAL SYSTEM

$$a + b = 10$$

$$2a + 2b = 20$$

$$5a + b = 11$$

$$10a + 2b = 22$$

$$0a + 0b = 0$$

$$0a + 0b = 0$$

ORIGINAL MATRIX

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



AFTER ELIMINATION

$$a + b = 10$$

$$0a + 0b = 0$$

$$a + 0.2b = 2.2$$

$$0a + 0b = 0$$

$$0a + 0b = 0$$

$$0a + 0b = 0$$

ROW ECHELON FORM

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

redundant

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

redundant

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

no information at all

A row of zeros appears in the row echelon form exactly when the system is singular.

Elementary row operations

DEFINITION

The three **elementary row operations** on a matrix are: swapping two rows, multiplying a row by a non-zero scalar, and adding a multiple of one row to another row.

Theorem. Elementary row operations do not change the solution set of the corresponding system, and they do not change whether the matrix is singular.

Swap

$$R_i \leftrightarrow R_j$$

Reorders the equations. Nothing about the solutions changes.

reversible

Scale

$$R_i \leftarrow k \cdot R_i, k \neq 0$$

Multiplies one equation by a constant. Dividing by k undoes it.

reversible

Replace

$$R_i \leftarrow R_i + k \cdot R_j$$

Adds a multiple of another equation. Subtracting $k \cdot R_j$ undoes it.

reversible

Row operations and the determinant

Start from the same matrix each time and watch what happens to $\det = 5 \cdot 3 - 1 \cdot 4 = 11$.

Switch the rows

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 3 \\ 5 & 1 \end{bmatrix}$$

$$\det = 11$$

$$\det = 4 \cdot 1 - 3 \cdot 5 = -11$$

The sign flips.

Multiply row 1 by 10

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 50 & 10 \\ 4 & 3 \end{bmatrix}$$

$$\det = 11$$

$$\det = 50 \cdot 3 - 10 \cdot 4 = 110 = 10 \cdot 11$$

The determinant is multiplied by 10.

Add row 1 to row 2

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 5 & 1 \\ 9 & 4 \end{bmatrix}$$

$$\det = 11$$

$$\det = 5 \cdot 4 - 1 \cdot 9 = 11$$

The determinant does not change.

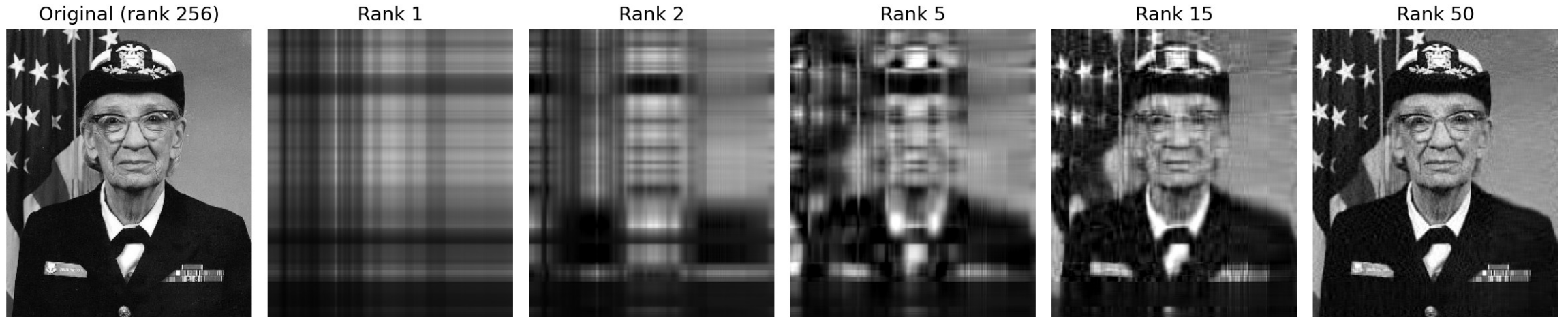
A determinant that is zero stays zero, and one that is non-zero stays non-zero: row operations preserve singularity.

PART 3

Rank of a matrix

Two equations do not always carry two pieces of information. Rank measures how much they really say.

Compressing images: reducing rank



A grayscale image is a matrix: one number per pixel. This one is 300×256 and has rank 256, so it carries 256 independent pieces of information.

Keeping only the k most important pieces (a rank- k approximation) rebuilds the picture from far fewer numbers: rank 1 is just stripes, rank 15 is recognisable, rank 50 is close to the original.

Rank is the amount of information in a matrix. The same idea powers image compression, principal component analysis and recommender systems.

Systems of information

System 1

The dog is black.
The cat is orange.

Two sentences

Two pieces of information

Rank = 2

System 2

The dog is black.
The dog is black.

Two sentences

One piece of information

Rank = 1

System 3

The dog.
The dog.

Two sentences

Zero pieces of information

Rank = 0

Rank counts independent pieces of information, not the number of statements.

Rank of a 2×2 system

System 1

$$a + b = 0$$

$$a + 2b = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Two equations

Two pieces of information

Rank = 2

System 2

$$a + b = 0$$

$$2a + 2b = 0$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Two equations

One piece of information

Rank = 1

System 3

$$0a + 0b = 0$$

$$0a + 0b = 0$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Two equations

Zero pieces of information

Rank = 0

The rank of the matrix is the rank of the system it represents.

Rank and the solution space

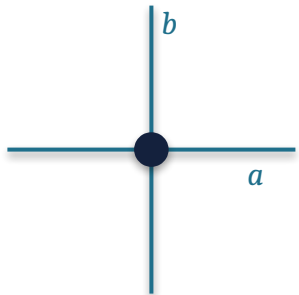
DEFINITION

For a homogeneous system (all constants equal to 0) the **solution space** is the set of all its solutions; its **dimension** is the number of free parameters needed to describe it. **Rank of the matrix = number of variables – dimension of the solution space.**

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Rank = 2

solution space has dimension 0

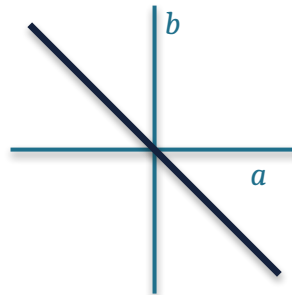


only the point (0, 0)

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Rank = 1

solution space has dimension 1

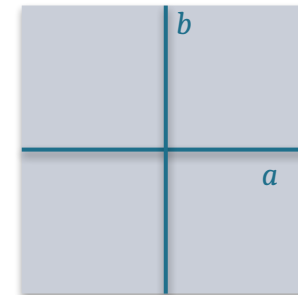


the line $b = -a$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank = 0

solution space has dimension 2



the whole plane

With two variables: Rank = 2 – (dimension of the solution space).

Rank and singularity

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Rank = 2

Non-singular

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Rank = 1

Singular

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank = 0

Singular

FACT

A square $n \times n$ matrix is non-singular exactly when its rank equals n (**full rank**). Any smaller rank means at least one row carries no new information: the matrix is singular.

Data science connection. If two feature columns are proportional (collinear), $X^T X$ loses rank and least squares has no unique solution.

Find the rank of each matrix

MATRIX 1

$$\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$$

MATRIX 2

$$\begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$$

Hint: write the homogeneous system (constants 0), find its solution space, and use

Rank = 2 – dimension of the solution space.

Quiz 3 · Solution

Matrix 1

$$\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$$

$$5a + b = 0$$

$$-a + 3b = 0$$

From the first equation $b = -5a$. Substituting: $-a + 3(-5a) = -16a = 0$, so $a = 0$ and then $b = 0$.

The solution space is the single point $(0, 0)$: dimension 0.

Rank = 2 - 0 = 2. Non-singular (det = 15 + 1 = 16 $\neq$ 0).

Matrix 2

$$\begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$$

$$2a - b = 0$$

$$-6a + 3b = 0$$

The second row is -3 times the first, so the second equation repeats the first.

Every point on the line $b = 2a$ is a solution: dimension 1.

Rank = 2 - 1 = 1. Singular (det = 6 - 6 = 0).

Rank of a 3×3 system

System 1

$$a + b + c = 0$$

$$a + 2b + c = 0$$

$$a + b + 2c = 0$$

3 equations

3 pieces of information

1	1	1
1	2	1
1	1	2

Rank 3

System 2

$$a + b + c = 0$$

$$a + b + 2c = 0$$

$$a + b + 3c = 0$$

3 equations

2 pieces of information

1	1	1
1	1	2
1	1	3

Rank 2

System 3

$$a + b + c = 0$$

$$2a + 2b + 2c = 0$$

$$3a + 3b + 3c = 0$$

3 equations

1 piece of information

1	1	1
2	2	2
3	3	3

Rank 1

System 4

$$0a + 0b + 0c = 0$$

$$0a + 0b + 0c = 0$$

$$0a + 0b + 0c = 0$$

3 equations

0 pieces of information

0	0	0
0	0	0
0	0	0

Rank 0

QUESTION

An easier way to compute the rank?

ANSWER

Yes. Bring the matrix to row echelon form and count the pivots. Equivalently, count the ones on the diagonal of its reduced row echelon form. The next part makes both forms precise.

PART 4

Row echelon form

A staircase shape that elimination always reaches, and the fastest way to read off the rank.

Row echelon form: definition

DEFINITION

A matrix is in **row echelon form** if

- (1) all rows consisting entirely of zeros are at the bottom;
- (2) in each non-zero row the first non-zero entry, called the **pivot**, lies strictly to the right of the pivot of the row above.

The **rank** of a matrix is the number of pivots in its row echelon form.

Note. In general pivots other than 1 are allowed. In this class we divide each row by its pivot so that every pivot is 1; this makes no mathematical difference.

2	*	*	*	*
0	1	*	*	*
0	0	3	*	*
0	0	0	-5	*
0	0	0	0	1

Rank 5

3	*	*	*	*
0	0	1	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

Rank 3

- Zero rows sit at the bottom.
- Each non-zero row has a pivot (its leftmost non-zero entry).
- Every pivot is to the right of the pivots above it: a staircase.
- The rank is the number of pivots: 5 and 3.

Reaching row echelon form, step by step



THE STEPS

Step 1 divide each row by its leftmost coefficient: $5a + b = 17$ becomes $a + 0.2b = 3.4$, $4a - 3b = 6$ becomes $a - 0.75b = 1.5$

Step 2 subtract row 1 from row 2: $\begin{bmatrix} 1 & -0.75 \end{bmatrix} - \begin{bmatrix} 1 & 0.2 \end{bmatrix} = \begin{bmatrix} 0 & -0.95 \end{bmatrix}$

Step 3 divide row 2 by its leftmost non-zero coefficient: $\begin{bmatrix} 0 & -0.95 \end{bmatrix} \div (-0.95) = \begin{bmatrix} 0 & 1 \end{bmatrix}$

Row echelon form, singularity and rank

Non-singular matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

2 pivots

in the diagonal

Rank 2

Singular matrix

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

1 pivot

in the diagonal

Rank 1

Singular matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

0 pivots

in the diagonal

Rank 0

Full rank (2 pivots) means non-singular; a missing pivot means a zero row and a singular matrix.

Row echelon form of a 3×3 system

SYSTEM

$$a + b + 2c = 12$$

$$3a - 3b - c = 3$$

$$2a - b + 6c = 24$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 6 \end{bmatrix}$$

Matrix

$$\begin{aligned} R_2 &\leftarrow R_2 - 3R_1 & R_3 &\leftarrow R_3 - 2R_1 \\ &\text{then } R_3 &\leftarrow 2R_3 - R_2 \end{aligned}$$



Three pivots: rank 3. The last row gives $c = 3$ at once, exactly as in the elimination we did by hand.

TRIANGULAR SYSTEM

$$a + b + 2c = 12$$

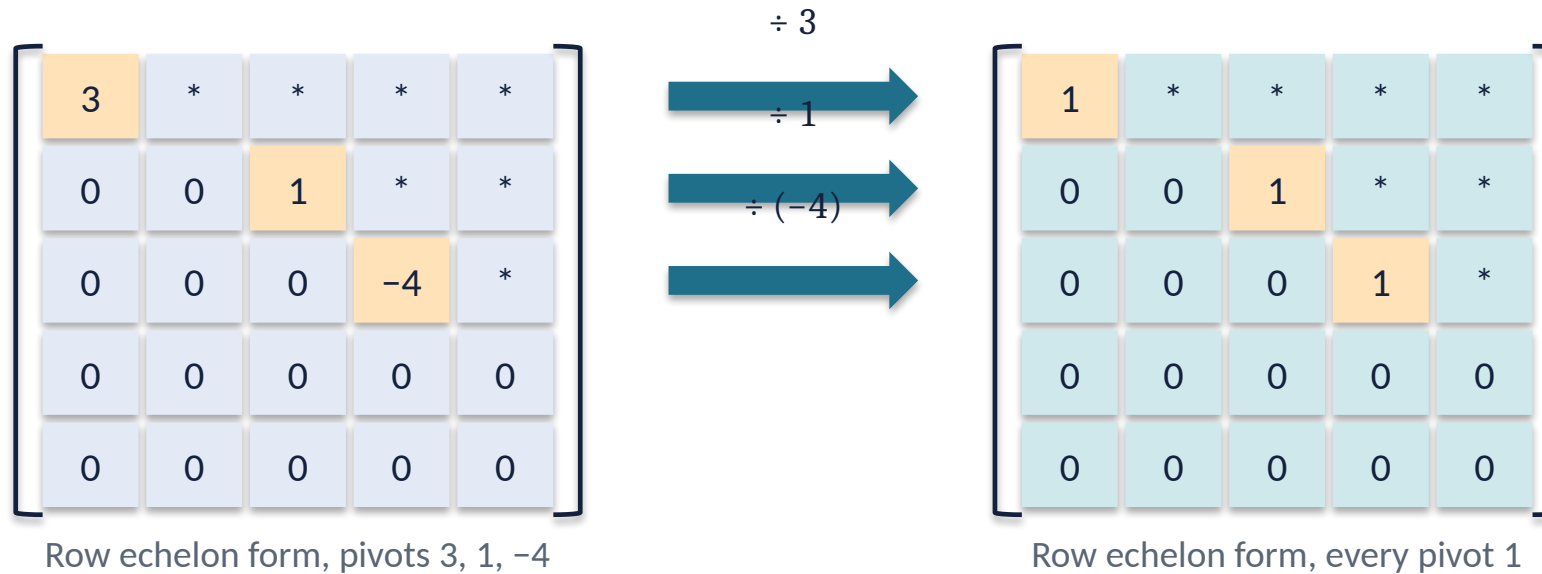
$$-6b - 7c = -33$$

$$11c = 33$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -6 & -7 \\ 0 & 0 & 11 \end{bmatrix}$$

Row echelon form matrix

Normalizing the pivots to 1



WHY IT IS ALLOWED

Dividing a row by its pivot is a legal row operation (scaling by a non-zero constant).

The staircase, the number of pivots and the rank are unchanged.

More examples

Non-singular

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - R_1}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

rank 3

Singular

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - R_1}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 2R_2}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

rank 2

Singular

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - 3R_1}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

rank 1

The second example needs two rounds: after the first round, row 3 is twice row 2.

Pivots count the rank

MATRIX

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

ROW ECHELON FORM

1	1	1
0	1	0
0	0	1

Number of pivots = 3

Rank = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

Rank = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

Rank = 1

0	0	0
0	0	0
0	0	0

Number of pivots = 0

Rank = 0

PART 5

Reduced row echelon form

Continue past the staircase: pivots become 1 and everything above a pivot becomes 0.

Reduced row echelon form: definition

DEFINITION

A matrix is in **reduced row echelon form** if

- (1) it is in row echelon form;
- (2) every pivot equals 1;
- (3) every pivot is the only non-zero entry in its column, so all entries above a pivot are 0.

Fact. Every matrix has exactly one reduced row echelon form; the rank is still the number of pivots.

For a non-singular square matrix the reduced row echelon form is the identity matrix, and the augmented system is then solved outright.

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

Rank 5

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Rank 3

- Is in row echelon form.
- Each pivot is a 1.
- Any number above a pivot is 0.
- Rank of the matrix is the number of pivots.

Reduced row echelon form: a 2×2 example

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Row echelon form



$$R_1 \leftarrow R_1 - 0.2 R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reduced row echelon form

The only entry above a pivot was the 0.2. One row operation clears it.

THE COMPUTATION

$$0.2 \times [0 \ 1] = [0 \ 0.2]$$

$$[1 \ 0.2] - [0 \ 0.2] = [1 \ 0] \quad \text{the entry above the second pivot is now 0}$$

Reduced row echelon form: the general recipe

3	*	*	*	*
0	0	2	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

Row echelon form



divide each row
by its pivot

1	*	*	*	*
0	0	1	*	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0



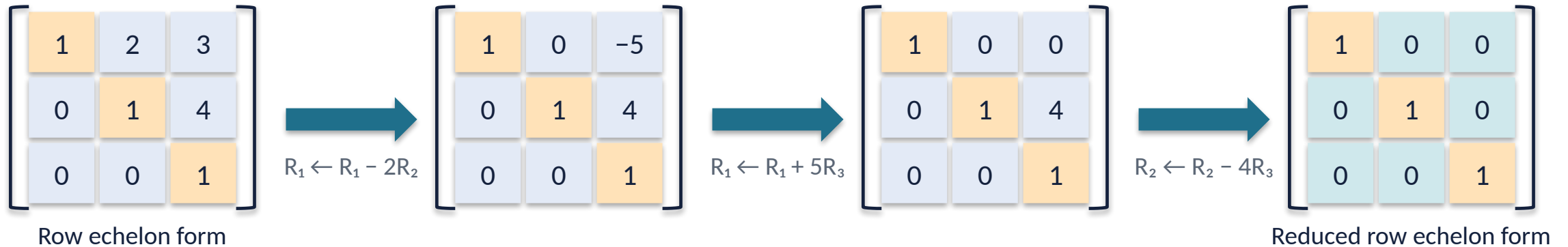
clear entries
above pivots

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Reduced row echelon form

Columns without a pivot keep their entries (the stars): they correspond to free variables of the system. The pivot columns become columns of the identity matrix.

Reduced row echelon form: a 3×3 example



Work from the last pivot upwards, or from the first pivot downwards: either order clears every entry above a pivot and never disturbs a column already finished. The result is the identity, so the matrix has rank 3 and is non-singular.

PART 6

The Gaussian elimination algorithm

Putting it all together on the augmented matrix: pivot, reduce, back-substitute, and read the answer.

The augmented matrix

DEFINITION

The **augmented matrix** $[A \mid b]$ of the system $A \cdot x = b$ is the coefficient matrix A with the column of constants b appended on the right. Row operations act on entire rows, constants included, so the system stays equivalent.

SYSTEM

$$2a - b + c = 1$$

$$2a + 2b + 4c = -2$$

$$4a + b = 4$$



AUGMENTED MATRIX

$$\begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ 2 & 2 & 4 & -2 \\ 4 & 1 & 0 & 4 \end{array} \right]$$

Each row is one equation; the bar separates the coefficients of a , b , c from the constant. Now proceed with the elimination method on the rows.

Pivoting on the first column

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ 2 & 2 & 4 & -2 \\ 4 & 1 & 0 & 4 \end{array} \right]$$

Start: pivot 2 in row 1



$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 2 & 2 & 4 & -2 \\ 4 & 1 & 0 & 4 \end{array} \right]$$

$R_1 \leftarrow \frac{1}{2} R_1$



$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 3 & 3 & -3 \\ 0 & 3 & -2 & 2 \end{array} \right]$$

$R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - 4R_1$

THE ROW ARITHMETIC

$$R_2 - 2R_1 : [2 \ 2 \ 4 \mid -2] - 2 \cdot [1 \ -\frac{1}{2} \ \frac{1}{2} \mid \frac{1}{2}] = [0 \ 3 \ 3 \mid -3]$$

$$R_3 - 4R_1 : [4 \ 1 \ 0 \mid 4] - 4 \cdot [1 \ -\frac{1}{2} \ \frac{1}{2} \mid \frac{1}{2}] = [0 \ 3 \ -2 \mid 2]$$

Pivoting on the second and third columns

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 3 & 3 & -3 \\ 0 & 3 & -2 & 2 \end{array} \right]$$

 $R_2 \leftarrow \frac{1}{3} R_2$


$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -5 & 5 \end{array} \right]$$

 $R_3 \leftarrow R_3 - 3R_2, \text{ then } R_3 \leftarrow -\frac{1}{5} R_3$


$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

Row echelon form

THE ROW ARITHMETIC

$$R_3 - 3R_2 : [0 \ 3 \ -2 \ | \ 2] - 3 \cdot [0 \ 1 \ 1 \ | \ -1] = [0 \ 0 \ -5 \ | \ 5]$$

$$-\frac{1}{5} R_3 : [0 \ 0 \ -5 \ | \ 5] \div (-5) = [0 \ 0 \ 1 \ | \ -1]$$

AS EQUATIONS

$$a - \frac{1}{2} b + \frac{1}{2} c = \frac{1}{2}$$

$$b + c = -1$$

$$c = -1$$

Back substitution

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

Row echelon form



$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

$R_2 \leftarrow R_2 - R_3, \quad R_1 \leftarrow R_1 - \frac{1}{2} R_3$



$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

$R_1 \leftarrow R_1 + \frac{1}{2} R_2$

THE RESULT

$$\mathbf{a} = 1, \quad \mathbf{b} = 0, \quad \mathbf{c} = -1$$

The coefficient part is the identity matrix; the constants column is the solution.

CHECK

$$2(1) - 0 + (-1) = 1 \quad \checkmark$$

$$2(1) + 2(0) + 4(-1) = -2 \quad \checkmark$$

$$4(1) + 0 = 4 \quad \checkmark$$

What if the system is singular?

Look at the row of zeros, then at its constant.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 2 & 4 & 5 & 1 \\ 3 & 6 & 4 & 6 \end{array} \right]$$

after row reduction

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 0 & 0 & 7 & -9 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$0a + 0b + 0c = 0$$

Infinitely many solutions

the constant in the zero row is 0

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 2 & 4 & 5 & 1 \\ 3 & 6 & 4 & 10 \end{array} \right]$$

after row reduction

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 0 & 0 & 7 & -9 \\ 0 & 0 & 0 & 4 \end{array} \right]$$

$$0a + 0b + 0c = 4$$

No solutions

the constant in the zero row is not 0

Row operations used: $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, $R_3 \leftarrow R_3 - R_2$. Only the constants column differs between the two systems.

Gaussian elimination: the algorithm

1 Create the augmented matrix

Write $[A \mid b]$: one row per equation, the constants in the last column.



2 Reach row echelon form

Pivot column by column: scale the pivot row to make the pivot 1, then subtract multiples of it from the rows below.



3 Reduce further or back-substitute

Clear the entries above each pivot to reach reduced row echelon form, or solve from the last equation upwards.



4 Stop at a row of zeros

A row $0 \dots 0 \mid 0$ means infinitely many solutions; a row $0 \dots 0 \mid \beta$ with $\beta \neq 0$ means no solution. Otherwise the solution is unique: the last column.

DEFINITION

Gaussian elimination is this procedure: elementary row operations applied to the augmented matrix until it is in (reduced) row echelon form.

Data science connection. This is what `numpy.linalg.solve` and R's `solve()` run under the hood (with row swaps chosen for numerical stability) whenever a model fits a linear system.

The definitions to remember

Linear system

A finite set of linear equations in the same variables; a solution satisfies all of them at once.

Singular vs non-singular

Non-singular: exactly one solution. Singular: infinitely many (redundant) or none (contradictory).

Row echelon form

Zero rows at the bottom; each pivot strictly to the right of the pivot above it.

Augmented matrix

$[A \mid b]$: the coefficient matrix with the constants column attached.

Equivalent systems

Same solutions. Swapping, scaling by $k \neq 0$, and adding a multiple of one equation to another keep a system equivalent.

Rank

The number of independent pieces of information; number of variables minus the dimension of the solution space; the number of pivots.

Reduced row echelon form

Row echelon form with every pivot equal to 1 and zeros above every pivot. Unique for each matrix.

Gaussian elimination

Row-reduce the augmented matrix to (reduced) row echelon form; read the solution or detect a zero row.